Containerization-and-DevOps

🐳 Class Task

🛠️ Tools Used

- Docker
- Nginx
- Ubuntu
- Alpine Linux

⚙️ Steps

Create Dockerfile (Ubuntu Base)

```
FROM ubuntu:22.04

RUN apt-get update && \
    apt-get install -y nginx && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Create Dockerfile (Alpine Base)

```
FROM alpine:latest

RUN apk add --no-cache nginx

EXPOSE 80

CMD ["nginx", "-g", "daemon off;"]
```

Create HTML File

```
<h1>Hello from Docker NGINX</h1>
```

Build Docker Image

```
docker build -t nginx-app .
```

Run Container

```
docker run -d -p 8080:80 nginx-app
```

Access Application

Open browser:

```
http://localhost:8080
```

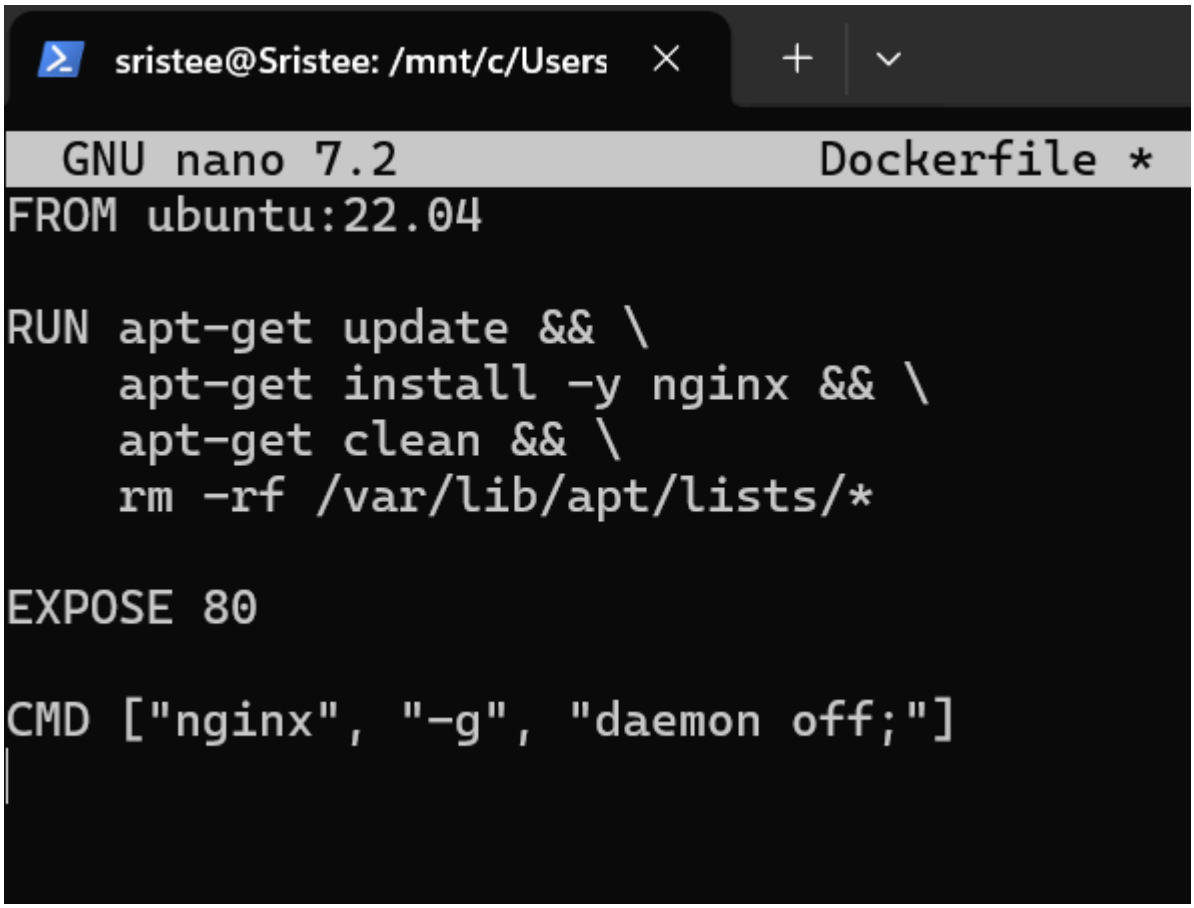
⚠️ Notes

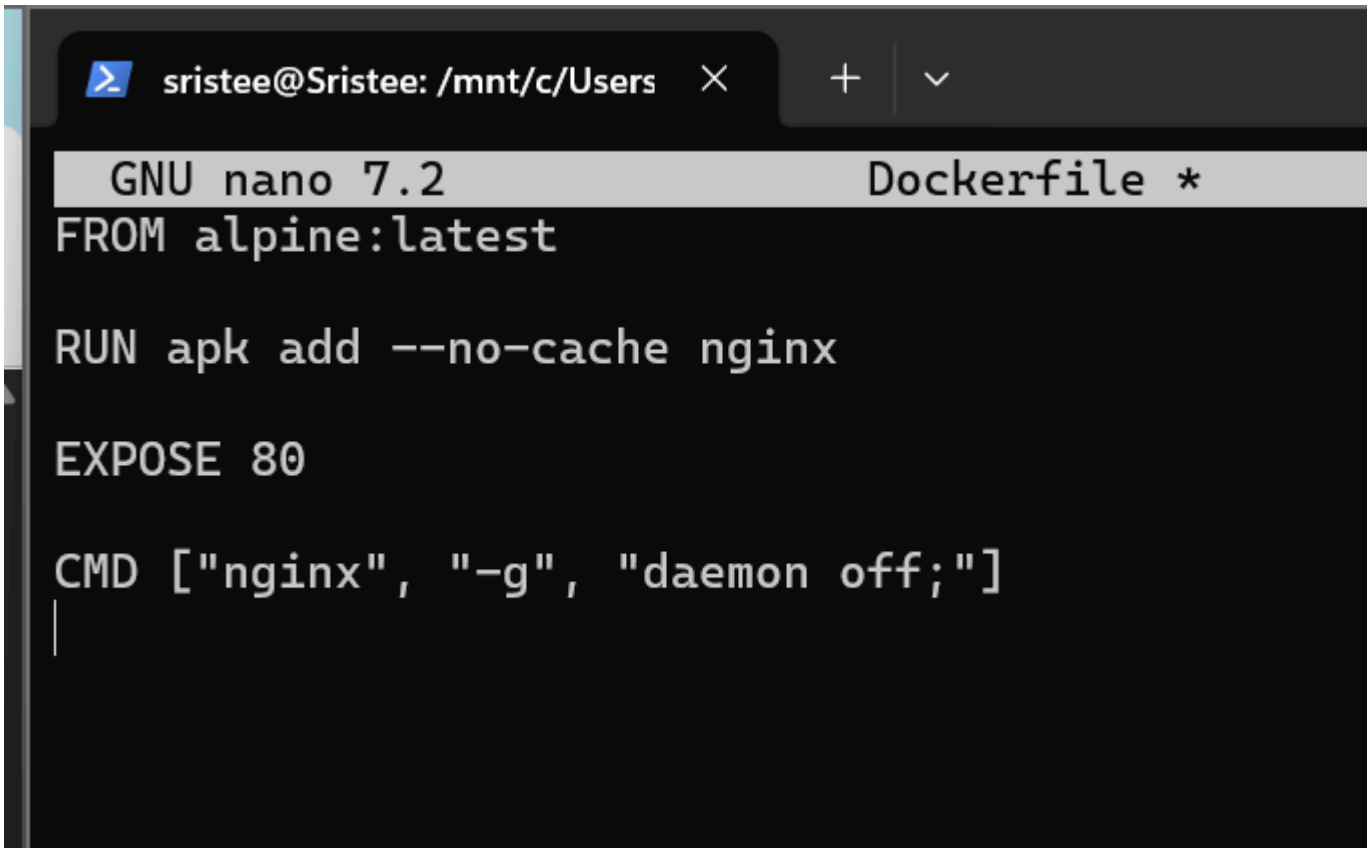
- Ubuntu image is larger but more familiar
- Alpine image is lightweight and faster
- Nginx runs in foreground using `daemon off;`
- Port 80 inside container mapped to 8080 on host

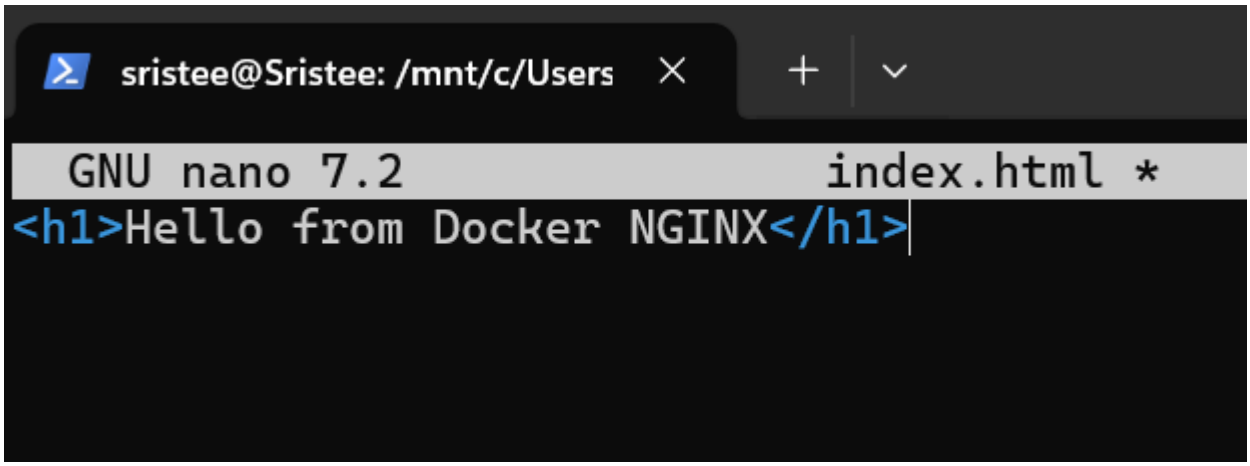
✅ Outcome

- Created Docker images using two base images
- Deployed Nginx container
- Served custom HTML page
- Compared Ubuntu vs Alpine images

Screenshots






sristee@Sristee: /mnt/c/Users
GNU nano 7.2 index.html *
<h1>Hello from Docker NGINX</h1>